

Test 14 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.DP.1.2 challenging	16	Add all the values, then divide by how many values there are.
2	MA.6.GR.2.2 challenging	120	The area of a parallelogram is base times height, using the height that is perpendicular to the base.
3	MA.6.AR.2.1 challenging	2	Substitute each value for x and check which one makes both sides equal.
4	MA.6.AR.3.1 challenging	9:2	Write bus students first, then divide both numbers by their greatest common factor.
5	MA.6.NSO.3.5 moderate	$\frac{3}{5}$	A percent means hundredths, so write it over 100 and simplify.
6	MA.6.GR.2.4 challenging	104	Add the areas of all six rectangles in the net; opposite faces match.
7	MA.6.NSO.3.2 moderate	64	Add inside the parentheses first, then multiply by the factor outside.
8	MA.6.NSO.2.2 moderate	$5 \frac{1}{5}$	Write the mixed number as a fraction, then multiply by the reciprocal of the divisor.
9	MA.6.DP.1.3 challenging	14	The median is shown by the vertical line inside the box.
10	MA.6.AR.2.4 challenging	1.7	Subtract the known addend from the total.
11	MA.6.NSO.4.2 easy	-24	A loss each week is a negative change repeated, so multiply.
12	MA.6.NSO.1.4 moderate	16	Take each absolute value first, then subtract.
13	MA.6.DP.1.5 challenging	2, $15 \frac{1}{2}$, 21, $28 \frac{1}{2}$, 37	Order the data, find the median, then find the medians of the lower and upper halves.
14	MA.6.GR.2.1 challenging	15	A triangle is half of a rectangle with the same base and height, so use one half times base times height.
15	MA.6.AR.3.4 challenging	16	40% means 40 out of every 100, so multiply the total by $\frac{40}{100}$.
16	MA.6.NSO.1.3 moderate	$2 \frac{3}{4}$	The distance from 0 is the absolute value of the number.
17	MA.6.DP.1.6 challenging	$19 \frac{1}{6}$	Add the new value to the total and divide by the new number of values.
18	MA.6.GR.2.3 challenging	210	Volume is length times width times height, or the area of the base times the height.
19	MA.6.AR.1.3 challenging	313	Evaluate the exponent before multiplying, then subtract.

#	Benchmark	Answer	Explanation
20	MA.6.NSO.2.1 moderate	8	Multiply both numbers by the same power of 10 so the divisor is a whole number, then divide.
21	MA.6.DP.1.1 challenging	How many text messages did each student send yesterday?	A statistical question expects different answers from different people, so it produces a set of numerical data.
22	MA.6.AR.3.5 challenging	432 inches	Each foot is 12 inches, so multiply the number of feet by 12.
23	MA.6.AR.3.3 challenging	20	Each row multiplies the first ratio by the same number, here 4.
24	MA.6.NSO.3.3 moderate	$\frac{1}{4}$	Square the numerator and square the denominator.
25	MA.6.NSO.4.1 moderate	-36	Subtracting a number is the same as adding its opposite.
26	MA.6.AR.1.4 challenging	$9x + 54$	Multiply the number outside the parentheses by each term inside.
27	MA.6.NSO.3.1 moderate	84	The next time they leave together is the least common multiple of the two intervals.
28	MA.6.NSO.1.1 moderate	$-\frac{3}{4}$	$-\frac{3}{4}$ is more than -1 and less than 0, so it sits between them on the number line.
29	MA.6.NSO.1.2 moderate	$8\frac{1}{2}$	The opposite of a number is the same distance from 0 but on the other side.
30	MA.6.AR.1.2 challenging	$x \leq 15$	"At least" and "no more than" include the boundary value; "more than" and "under" do not.
31	MA.6.GR.1.1 moderate	$(-3, -2), (-5, -4), (-1, -5)$	A point is in Quadrant III when both coordinates are negative.
32	MA.6.AR.2.3 challenging	3	Divide both sides by the coefficient to undo the multiplication.
33	MA.6.NSO.4.2 moderate	4	Divide the absolute values, then decide the sign from the two signs.
34	MA.6.AR.3.2 challenging	\$5.00	Divide the total cost by the number of pounds to get the cost for one pound.
35	MA.6.DP.1.4 challenging	7	Count one dot for each student at 5 and to the right of 5.
36	MA.6.NSO.3.4 moderate	90	Multiply all of the prime factors together.
37	MA.6.NSO.2.3 moderate	\$82.96	Find the cost of all the items first, then subtract that from the amount paid.
38	MA.6.AR.2.2 challenging	37	Subtract the same number from both sides to undo the addition.